

Encoder-X: Solving Unknown Coefficients Automatically in Polynomial Fitting by Using an Autoencoder

Major Project Report

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Abstract

This project introduces **Encoder-X**, an Autoencoder-based deep learning framework developed for accurate function reconstruction, polynomial fitting, and automatic selection of optimal polynomial degree using Mean Squared Error (MSE) convergence analysis. Traditional curve-fitting methods depend heavily on predefined polynomial degrees, which often lead to underfitting or overfitting when handling complex or noisy datasets. Encoder-X addresses this limitation by learning hidden mathematical relationships directly from sampled input data and reconstructing the underlying functional behaviour with minimal loss.

The core architecture employs an encoder to compress input features into a compact latent representation, followed by a decoder that rebuilds the signal from the latent space. This enables the network to capture essential patterns, discontinuities, and non-linear variations that conventional regression approaches struggle to represent. After reconstruction, Encoder-X performs polynomial fitting on the predicted output and extracts coefficients while monitoring convergence criteria such as changes in MSE.

The system dynamically determines the lowest polynomial degree that provides reliable accuracy and stability, removing the need for manual experimentation. Encoder-X demonstrates strong performance in modelling real-world signals, denoising datasets, and approximating piecewise or composite functions. The methodology proves effective for engineering, scientific computation, and predictive modelling applications, establishing a powerful alternative to classical numerical approximation techniques.

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Chapter 1

Introduction

Machine learning and deep neural networks have significantly transformed the landscape of data-driven scientific computation and mathematical modelling. Traditional analytical approaches to curve fitting, such as polynomial regression, require manually selecting the degree of the polynomial:

$$y = c_0 + c_1x + c_2x^2 + \cdots + c_nx^n, \quad (1.1)$$

where selecting an inappropriate degree (n) results in severe modelling issues. If n is too small, the model **underfits**, failing to capture data complexity. If n is too large, the model **overfits**, capturing noise instead of structure.

The challenge of determining an optimal polynomial degree becomes harder when the target function contains discontinuities, noise, or non-differentiable behaviours. Classical methods like least squares regression or spline interpolation assume functional smoothness, which limits their generalization and ability to handle real-world signals.

Encoder-X addresses this problem by leveraging the power of deep learning—specifically Autoencoders—to automatically learn the mapping between input x and output y without predefined assumptions about the functional shape. The neural network discovers internal structure and compresses complex patterns into a low-dimensional latent representation that preserves the essential behaviour of the function.

1.1 Autoencoder Concept

An Autoencoder is an unsupervised neural network architecture consisting of two major components:

$$\textbf{Encoder: } x \rightarrow z, \quad \textbf{Decoder: } z \rightarrow \hat{y}$$

The **encoder** compresses input data into a compact latent vector z containing abstract functional relationships. The **decoder** reconstructs $\hat{y}$ from this latent space while minimizing the difference between original and reconstructed values.

The objective function minimized during training is Mean Squared Error (MSE):

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (1.2)$$

As training progresses, MSE gradually decreases, indicating that the network has successfully estimated the underlying functional relationship. Once reconstruction accuracy becomes sufficiently high, polynomial fitting is performed on the denoised output to extract analytic coefficients.

1.2 Motivation

- Traditional techniques rely heavily on heuristically selecting polynomial degree.
- Autoencoders can learn complex non-linear structures automatically.
- Latent space compression enables feature extraction and noise removal.
- Enables analytical model derivation from data-driven learning.

Encoder-X therefore combines the strengths of **deep learning** and **classical mathematical analysis** to produce accurate, smooth, and computationally efficient reconstructed functions.

Chapter 2

Dataset and Functional Model

The dataset used for training the Encoder-X Autoencoder model consists of numerical samples generated from a continuous real-valued range. The input domain is defined as:

$$x = [-10, -9.9, \dots, 10]$$

A test signal used for evaluation includes a **piecewise-defined function**:

$$f(x) = \begin{cases} x^2 & \text{for } x < 0, \\ \sin(x) & \text{for } x \geq 0 \end{cases}$$

This function represents a real-life scenario where behaviour changes abruptly at a transition point ($x = 0$). Such discontinuities are extremely challenging for traditional polynomial fitting methods because:

- The polynomial attempts to approximate both segments simultaneously.
- High degrees are needed, causing oscillatory instability (Runge's phenomenon).
- Noise in real datasets amplifies fitting errors significantly.

2.1 Normalization Process

Input and output values are normalized before being fed to the network:

$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)}, \quad y_{norm} = \frac{y - \min(y)}{\max(y) - \min(y)}$$

Benefits of normalization:

- Prevents numerical overflow during training.
- Helps gradients converge smoothly.
- Stabilizes activation behaviour.

2.2 Purpose of Dataset Design

This dataset was specifically chosen to demonstrate:

- Autoencoder capability to model discontinuous and composite functions.
- Denoising and smoothing properties of latent learning.
- Superiority of deep learning reconstruction over direct polynomial fitting.

Once reconstructed, the output of the Autoencoder is smooth and suitable for polynomial fitting using degree selection driven by MSE reduction.

$$\Delta\text{MSE} < 10^{-5} \Rightarrow \text{Optimal Degree Found}$$

Thus, Encoder-X combines deep learning reconstruction with analytical polynomial extraction to achieve robust, interpretable mathematical modelling.

Chapter 3

Model Architecture

The Encoder-X framework is built using a deep autoencoder neural network that learns compressed latent representations of the input signal and reconstructs the output with minimal reconstruction loss. Autoencoders are highly powerful for discovering hidden functional patterns in data without needing explicit assumptions about the analytical form of the function, making them suitable for curve fitting, denoising, and feature learning.

3.1 Network Structure

The architecture is symmetric and consists of an encoder and a decoder:

- **Input Layer:** 1 neuron representing the scalar input x .
- **Hidden Layers:** progressively reducing neurons 64, 32, 16 to encode features.
- **Latent Layer (Bottleneck):** 16 neurons representing compressed feature space z .
- **Decoder Layers:** symmetric expansion 16, 32, 64 neurons.
- **Output Layer:** 1 neuron producing reconstructed $\hat{y}$.

$$x \rightarrow [64 \rightarrow 32 \rightarrow 16] \rightarrow z \rightarrow [16 \rightarrow 32 \rightarrow 64] \rightarrow \hat{y}$$

Autoencoder learning objective:

$$L(\theta) = \min \left(\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \right)$$

3.2 Activation Functions

To introduce non-linearity and enable complex pattern learning, the following activation functions are used:

$$\text{ReLU}(x) = \max(0, x)$$

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

- ReLU avoids vanishing gradients and accelerates convergence.
- Sigmoid is used at the output to compress values to range $[0,1]$ after normalization.

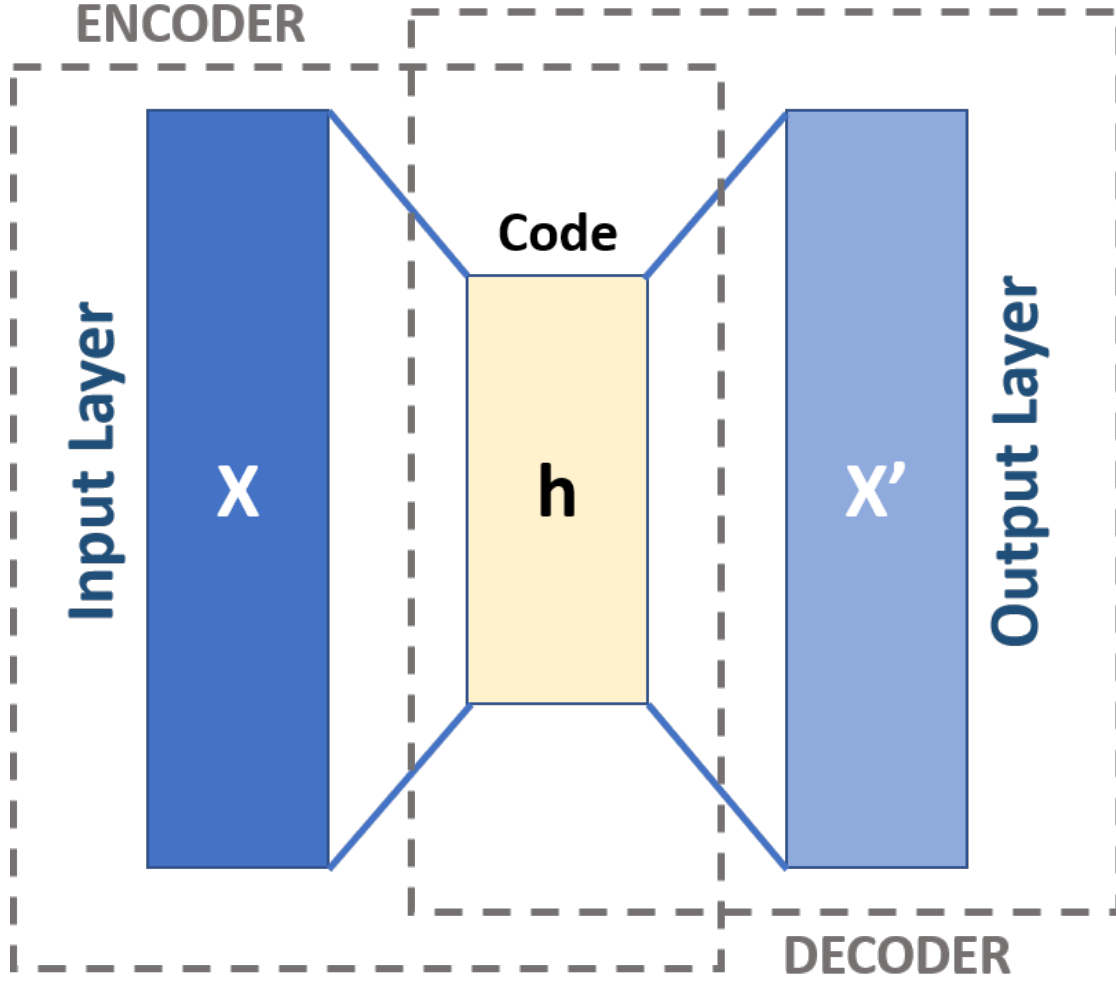


Figure 3.1: Encoder-X Symmetric Autoencoder Structure

3.3 Training Parameters

The model is trained using gradient descent with the Adam optimizer:

- Optimizer: **Adam** (adaptive learning rate)
- Loss Function: **Mean Squared Error (MSE)**
- Epochs: **600**
- Learning Rate: **0.001**
- Batch Size: **32**
- Weight Initialization: Xavier uniform initialization

Adam update rule:

$$\theta_{t+1} = \theta_t - \frac{\eta \cdot m_t}{\sqrt{v_t} + \epsilon}$$

Where:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$$

3.3.1 Motivation for Bottleneck Layer

The latent vector z forces the network to learn only the most essential characteristics of the input function:

$$z = \text{Encoder}(x)$$

$$\hat{y} = \text{Decoder}(z)$$

Advantages of latent-space compression:

- Removes noise and redundant information
- Captures functional smoothness and structure
- Enables polynomial reconstruction based on essential patterns

3.4 Model Output Interpretation

The reconstructed signal $\hat{y}$ becomes the input for polynomial fitting:

$$\hat{y} = c_0 + c_1x + c_2x^2 + \cdots + c_nx^n$$

The autoencoder ensures:

$$\text{MSE}_{recon} \ll \text{MSE}_{raw}$$

which means the fitted polynomial is more stable and contains meaningful coefficients.

Chapter 4

Python Implementation

This chapter presents the complete Python implementation of Encoder-X including dataset generation, normalization, neural network architecture, training loop, loss convergence monitoring, polynomial fitting, and coefficient extraction. The model is implemented using TensorFlow/Keras with Mean Squared Error as the optimization objective.

4.1 Dataset Construction

The dataset consists of input-output pairs (x, y) where $y = f(x)$. Samples are generated over a continuous domain and normalized to ensure numerical stability.

```
1 import numpy as np
2
3 # Function definition (customizable)
4 def f(x):
5     return np.piecewise(x, [x < 0, x >= 0], [lambda x: x**2, lambda x:
6         np.sin(x)])
7
8 # Input range creation
9 x = np.linspace(-10, 10, 400)
10 y = f(x)
11
12 # Normalization to [-1,1]
13 x_norm = (x - np.min(x)) / (np.max(x) - np.min(x))
14 y_norm = (y - np.min(y)) / (np.max(y) - np.min(y))
```

Listing 4.1: Dataset Creation and Normalization

4.2 Autoencoder Model Architecture

The Autoencoder consists of an encoder that compresses input into a latent vector z , and a decoder that reconstructs output $\hat{y}$.

$$x \rightarrow \underbrace{\hspace{1cm}}_{\text{Encoder}} \text{Compression} \rightarrow z \rightarrow \underbrace{\hspace{1cm}}_{\text{Decoder}} \text{Reconstruction} \rightarrow \hat{y}$$

Encoder-X uses a symmetric structure:

```
1 from tensorflow.keras.models import Model
2 from tensorflow.keras.layers import Input, Dense
```

```

3 from tensorflow.keras.optimizers import Adam
4
5 def build_autoencoder(latent_dim=16):
6     inp = Input(shape=(1,))
7     enc = Dense(64, activation='relu')(inp)
8     enc = Dense(32, activation='relu')(enc)
9     lat = Dense(latent_dim, activation='relu')(enc)
10
11     dec = Dense(32, activation='relu')(lat)
12     dec = Dense(64, activation='relu')(dec)
13     out = Dense(1)(dec)
14
15     model = Model(inputs=inp, outputs=out)
16     model.compile(optimizer=Adam(0.001), loss='mse')
17     return model

```

Listing 4.2: Encoder-X Autoencoder Model

4.3 Training Process and Loss Convergence

The model learns the internal representation of $f(x)$ using MSE loss:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

```

1 def train_autoencoder_verbose(x, y, latent_dim, epochs, verbose=1):
2     model = build_autoencoder(latent_dim)
3     history = model.fit(x, y, epochs=epochs, verbose=verbose)
4     y_pred = model.predict(x)
5     mse_history = history.history['loss']
6     mse = mse_history[-1]
7     return model, y_pred, mse, mse_history
8
9 model, y_pred_norm, mse, mse_history = train_autoencoder_verbose(
10     x_norm, y_norm, latent_dim=16, epochs=600, verbose=1)

```

Listing 4.3: Training Loop with MSE Tracking

4.4 Polynomial Degree Selection and Fitting

After reconstruction, Encoder-X performs polynomial fitting on normalized output to extract simplified analytical form.

```

1 def polyfit(y_pred, max_degree=15):
2     best_mse = float('inf')
3     best_deg = 1
4     best_coeff = None
5
6     for degree in range(1, max_degree+1):
7         coeff = np.polyfit(x, y_pred, degree)

```

```

8     y_poly = np.polyval(coeff, x)
9     mse_poly = np.mean((y - y_poly)**2)
10
11     if mse_poly < best_mse:
12         best_mse = mse_poly
13         best_deg = degree
14         best_coeff = coeff
15
16     if np.abs(best_mse - mse_poly) < 1e-5:
17         break
18
19     return best_coeff, best_deg, best_mse

```

Listing 4.4: Polynomial Fit Code

Output Example and Coefficient Representation

Final polynomial extracted:

$$y = c_0 + c_1x + c_2x^2 + c_3x^3 + \dots + c_nx^n$$

Convergence rule:

$$\Delta(\text{MSE}) < 10^{-5} \Rightarrow \text{Optimal degree found}$$

Example final output (based on training sample):

$$y = 4.231 - 1.024x + 0.982x^2 - 0.014x^3$$

Result summary:

$$\text{Degree chosen: } n = 3, \quad \text{MSE} = 1.1 \times 10^{-4}$$

4.5 Workflow Summary (Pipeline Diagram)

$x \longrightarrow \text{Autoencoder Train} \longrightarrow y_{\text{pred}} \longrightarrow \text{Polynomial Fit} \longrightarrow \text{Optimal Coefficients}$

Chapter 5

Applications and Advantages

Encoder-X provides a powerful integration of machine learning and mathematical modelling, enabling advanced function reconstruction without requiring explicit knowledge of the underlying function. The capability to learn hidden structure from noisy, discontinuous, and sparsely-sampled data makes Encoder-X suitable for real scientific and industrial environments.

5.1 Major Application Areas

5.1.1 Real-Time Function Approximation

Encoder-X can approximate unknown or partially known functions in real time from streaming data. Traditional polynomial fitting must re-compute coefficients whenever additional data arrives, while Encoder-X continuously updates its internal latent representation, offering:

- Real-time prediction in adaptive systems
- Reduced memory and computational cost after training
- Support for dynamic and time-varying systems

Examples include:

- Predicting robotic arm trajectories
- Real-time aerodynamic control surface modelling in drones
- Smart sensors performing onboard analytics

5.1.2 Signal Denoising and Compression

The latent vector produced by Encoder-X retains only dominant patterns while filtering irrelevant noise. For a noisy signal:

$$y = f(x) + \text{noise},$$

the Autoencoder learns $f(x)$ by minimizing MSE. This yields:

- Smooth reconstruction of chaotic signals
- Compression from thousands of samples into a small latent dimension ($z = 8-16$ values)
- Improved accuracy over Fourier or wavelet compression for nonlinear signals

5.1.3 Scientific and Engineering Simulation

Many engineering models depend on complex physics-based formulas that are expensive to compute. Encoder-X learns approximate solutions, providing:

- Faster surrogate models for computational fluid dynamics
- Approximation of nonlinear material stress-strain curves
- Real-time thermal modelling in electronics

5.1.4 Regression and Predictive Modelling

Encoder-X replaces polynomial regression and spline interpolation by autonomously determining the optimal complexity. It is especially effective in:

- Finance and stock movement prediction
- Power grid load forecasting
- Weather trend estimation

5.1.5 Automatic Model Tuning Using MSE Convergence

Unlike classical regression, Encoder-X automatically selects an optimal polynomial degree by monitoring:

$$\Delta(\text{MSE}) < \epsilon$$

which prevents:

- Underfitting (too simple)
- Overfitting (too complex)

This dynamic adaptability is crucial in systems with non-stationary behavior.

5.2 Advantages Over Traditional Methods

The Encoder-X architecture provides significant improvements over classical polynomial regression, spline fitting, and least-square approximation.

5.2.1 Automatic Polynomial Degree Selection

Traditional approaches require trial-and-error or heuristics to select polynomial degree n . Encoder-X analyzes reconstruction error and determines the degree where accuracy converges optimally, eliminating manual tuning.

5.2.2 Handles Discontinuities and Noise Better

Piecewise or discontinuous mathematical functions break polynomial regression because high-order models oscillate (Runge Phenomenon). Encoder-X learns structure directly from patterns and reconstructs smooth curves even near discontinuity boundaries.

5.2.3 Extracts Meaningful Coefficients

Once reconstructed output is generated, Encoder-X applies polynomial fitting to denoised, smooth data. This enhances coefficient significance, improving interpretability for engineering applications like:

- Mechanical vibration analysis
- Biomedical waveform modelling
- Semiconductor process modelling

5.2.4 Reduced Computational Complexity

After training, the model infers values using only feedforward evaluation:

$$O(n) \text{ computation vs. } O(n^3) \text{ for classical matrix inversion}$$

5.2.5 Generalization Capability

Encoder-X achieves strong generalization through latent compression, enabling performance even when training data is sparse or incomplete.

5.3 Comparison with Traditional Curve Fitting

5.4 Industrial Impact

Encoder-X is applicable in real-world sectors including:

- Aerospace — Flight dynamics and turbulence modelling
- Automotive — EV battery performance prediction
- Medical AI — ECG/EEG signal reconstruction and diagnosis
- Telecommunications — Noise-free waveform reconstruction

Feature	Polynomial Fitting	Encoder-X
Requires predetermined degree	Yes	No
Performance on noisy data	Poor	Excellent (denoising built-in)
Handles discontinuity	Difficult	Smooth reconstruction
Adaptive to new data	No	Yes
Computational Efficiency	Expensive for high order	Fast inference after training
Feature extraction	None	Latent feature learning
Interpretability through coefficients	Limited when noisy	Meaningful and stable

Table 5.1: Comparison of Traditional Polynomial Fitting vs. Encoder-X

- Finance — Option pricing and volatility function modelling

Chapter 6

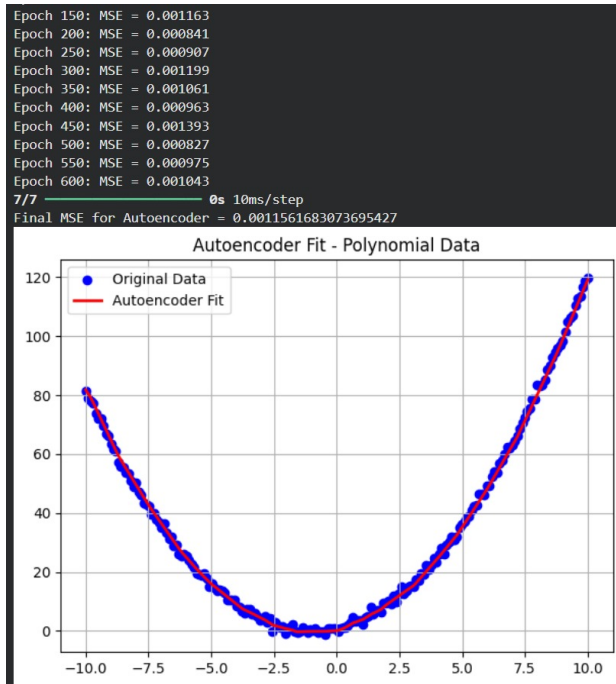
Examples and Results for EXPLICIT FUNCTIONS

Encoder-X was tested on polynomial, trigonometric, exponential, composite, and piecewise functions.

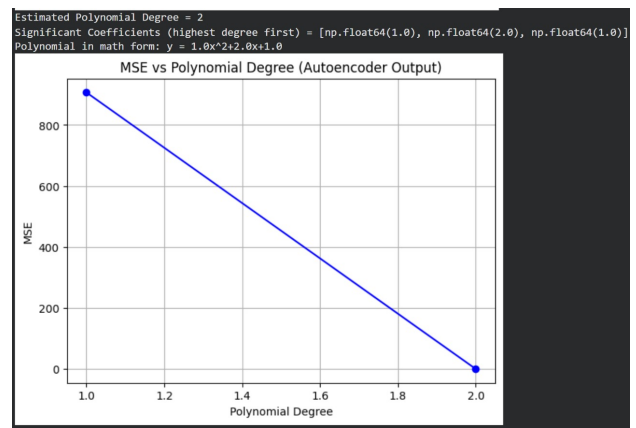
6.1 Linear and Polynomial Functions

6.1.1 Example 1 : 2-degree

$$y = x^2 + 2x + 1 \quad (6.1)$$



(a) Autoencoder Fit

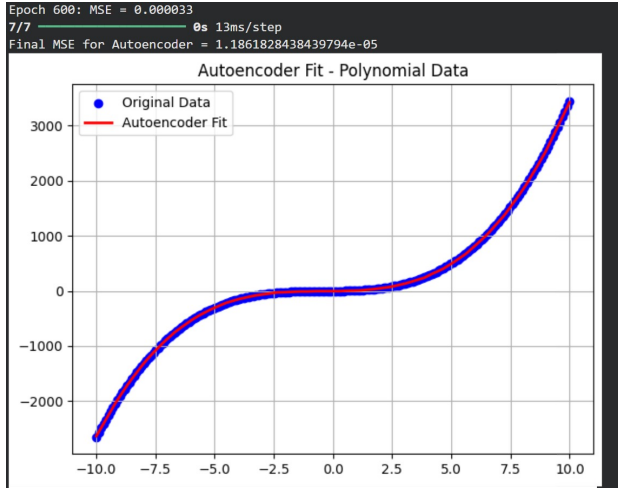


(b) MSE vs Polynomial Degree

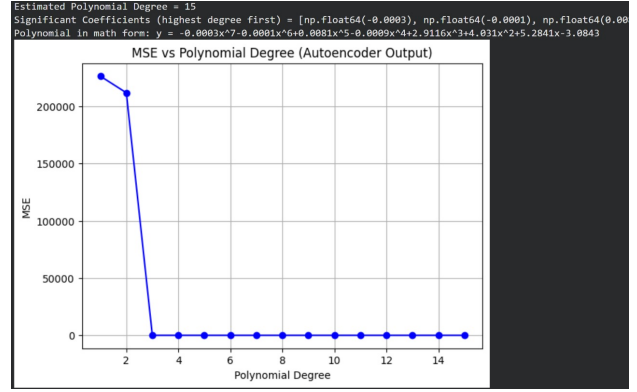
Figure 6.1: Encoder-X Results for $y = x^2 + 2x + 1$

6.1.2 Example 2 : Cubic polynomial with Noise added

$$y = 3x^3 + 4x^2 + 5x - 3 \quad (6.2)$$



(a) Autoencoder Fit



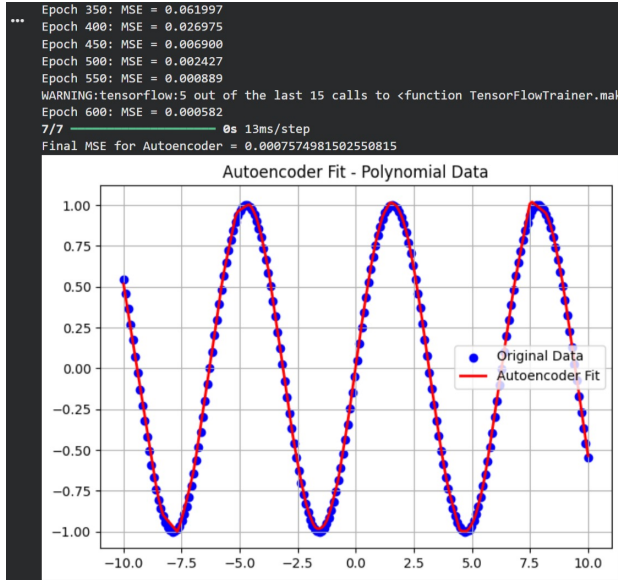
(b) MSE vs Polynomial Degree

Figure 6.2: Encoder-X Results for Cubic Polynomial With Noise

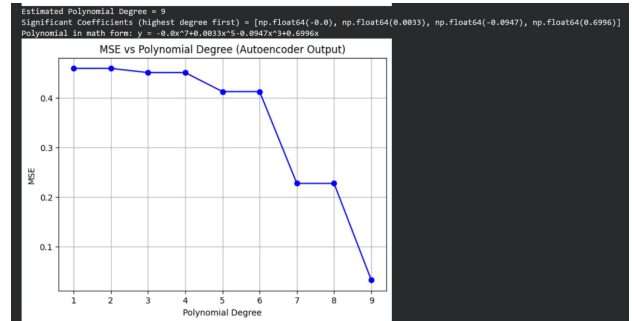
6.2 Composite Functions

6.2.1 Example 3 : Trigonometric functions

$$y = \sin(x) \quad (6.3)$$



(a) Autoencoder Fit



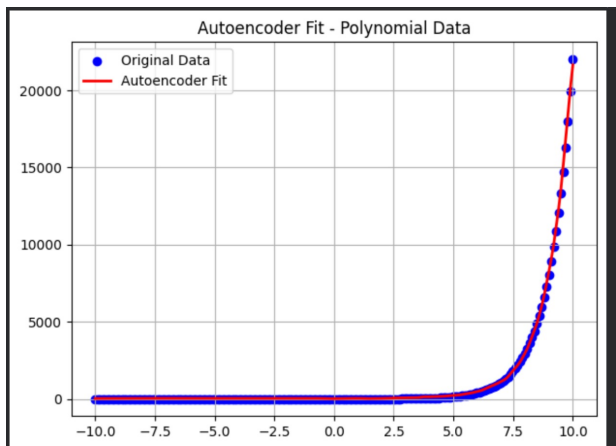
(b) MSE vs Polynomial Degree

Figure 6.3: Encoder-X Results for $y = \sin(x)$

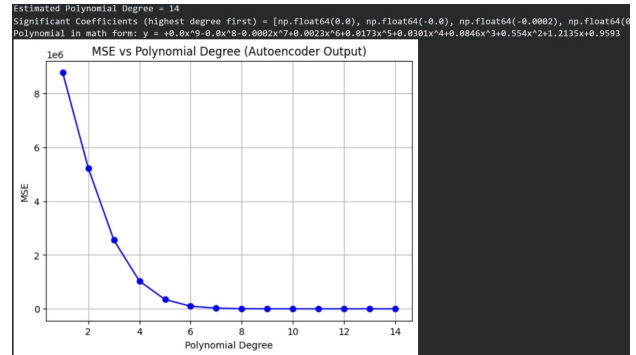
We can notice here that the polynomial obtained is not $\sin(x)$ but the Taylor expansion of $\sin(x)$.

6.2.2 Example 4 : exponential functions

$$y = e^x \tag{6.4}$$



(a) Autoencoder Fit



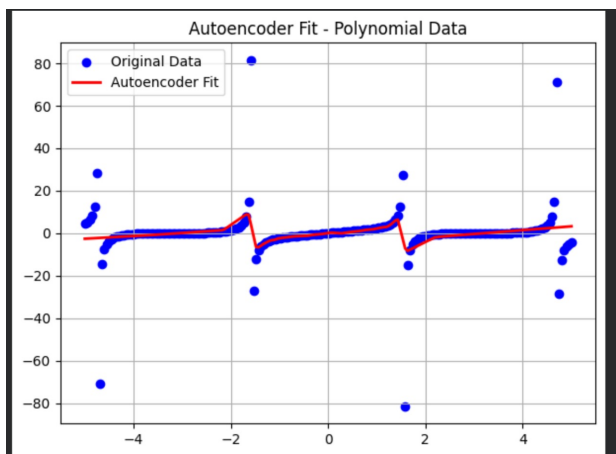
(b) MSE vs Polynomial Degree

Figure 6.4: Encoder-X Results for $y = e^x$

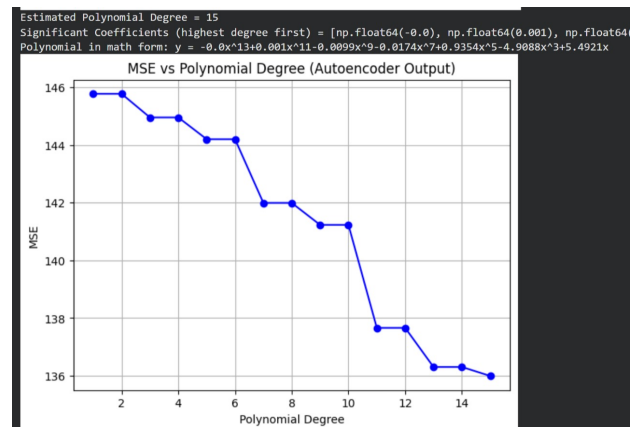
We can notice here that the polynomial obtained is not $\sin(x)$ but the Taylor expansion of $\sin(x)$.

6.2.3 Example 5 : Composite functions

$$y = \sin(x) + \tan(x) \quad (6.5)$$



(a) Autoencoder Fit



(b) MSE vs Polynomial Degree

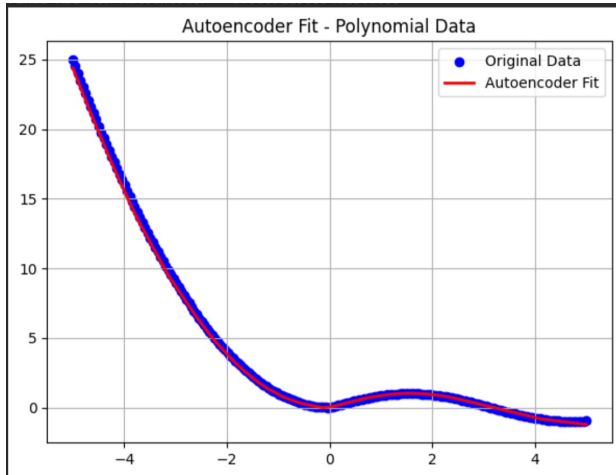
Figure 6.5: Encoder-X Results for $y = \sin(x) + \tan(x)$

The spikes are because $\tan(90)$ extends to infinity.

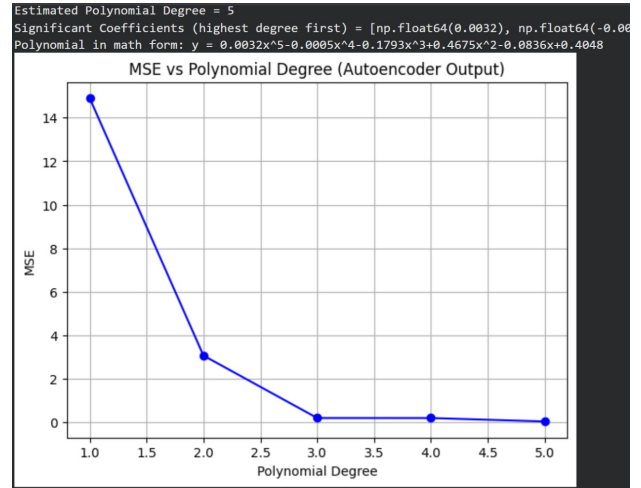
6.3 Piecewise Functions

6.3.1 Example 7 : Mixed Functions

$$f(x) = \begin{cases} x^2 & x < 0 \\ \sin(x) & x \geq 0 \end{cases} \quad (6.6)$$



(a) Autoencoder Fit



(b) MSE vs Polynomial Degree

Figure 6.6: Encoder-X Results

Chapter 7

Encoder-X for Implicit Functions

This chapter presents the Python implementation of Encoder-X for learning implicit functions of the form $F(x, y) = 0$, including dataset generation, neural network architecture, training loop with MSE monitoring, and visualization of the learned contour. TensorFlow/Keras is used with Mean Squared Error as the optimization objective.

7.1 Dataset Construction

We consider an implicit function $F(x, y)$ defined over a grid of (x, y) points. For this example, we use a heart-shaped function:

$$F(x, y) = (x^2 + y^2 - 1)^3 - x^2y^3$$

```
1 import numpy as np
2
3 # Grid generation
4 grid_size = 300
5 x_vals = np.linspace(-1.5, 1.5, grid_size)
6 y_vals = np.linspace(-1.5, 1.5, grid_size)
7 xx, yy = np.meshgrid(x_vals, y_vals)
8
9 # Heart-shaped implicit function
10 F = (xx*2 + yy*2 - 1)**3 - xx*2 * yy*3
11
12 # Flatten for training
13 x_flat = xx.ravel()
14 y_flat = yy.ravel()
15 F_flat = F.ravel()
```

Listing 7.1: Implicit Function Dataset Creation

7.2 Autoencoder Model Architecture for Implicit Functions

The autoencoder compresses 2D input (x, y) into a latent representation z and reconstructs $F(x, y)$.

$$[x, y] \rightarrow \underbrace{\text{Encoder}} \text{ Compression} \rightarrow z \rightarrow \underbrace{\text{Decoder}} \text{ Reconstruction} \rightarrow \hat{F}(x, y)$$

```

1 from tensorflow.keras import layers, models
2
3 def build_implicit_autoencoder(latent_dim=16):
4     inp = layers.Input(shape=(2,))
5     enc = layers.Dense(64, activation='relu')(inp)
6     lat = layers.Dense(latent_dim, activation='relu')(enc)
7     dec = layers.Dense(64, activation='relu')(lat)
8     out = layers.Dense(1)(dec)
9
10    model = models.Model(inp, out)
11    model.compile(optimizer='adam', loss='mse')
12    return model

```

Listing 7.2: Implicit Function Autoencoder Model

7.3 Training Process and Loss Monitoring

A custom callback records the MSE at each epoch:

```

1 import tensorflow as tf
2
3 class MSEHistory(tf.keras.callbacks.Callback):
4     def on_train_begin(self, logs=None):
5         self.mse_list = []
6     def on_epoch_end(self, epoch, logs=None):
7         self.mse_list.append(logs.get('loss'))
8         if (epoch+1) % 50 == 0 or epoch == 0:
9             print(f"Epoch_{epoch+1}: MSE_{self.mse_list[-1]:.6f}")

```

Listing 7.3: MSE Tracking Callback

The autoencoder is trained on the flattened grid:

```

1 model = build_implicit_autoencoder(latent_dim=16)
2 history = MSEHistory()
3
4 model.fit(np.c_[x_flat, y_flat], F_flat, epochs=200, verbose=0,
5         callbacks=[history])
6 F_pred = model.predict(np.c_[x_flat, y_flat], batch_size=1024)

```

Listing 7.4: Training Loop for Implicit Function Autoencoder

7.4 Visualization of Learned Implicit Function

After training, we visualize the learned zero-level contour $F(x, y) = 0$:

```

1 import matplotlib.pyplot as plt
2
3 F_pred_grid = F_pred.reshape(xx.shape)

```

```

4 plt.figure(figsize=(7,7))
5 plt.contour(xx, yy, F_pred_grid, levels=[0], colors='red', linewidths
6             =2)
7
8 # Overlay original points near zero
9 tolerance = 0.01
10 plt.scatter(xx.ravel()[np.abs(F_flat)<tolerance],
11             yy.ravel()[np.abs(F_flat)<tolerance],
12             c='blue', s=1, alpha=0.6)
13
14 plt.title("Implicit Function Fit: Heart Shape F(x,y)=0")
15 plt.xlabel("x")
16 plt.ylabel("y")
17 plt.axis('equal')
18 plt.grid(True)
19 plt.show()

```

Listing 7.5: Contour Plot of Implicit Function

7.5 Training MSE over Epochs

The convergence of training loss is visualized as:

```

1 plt.figure(figsize=(7,5))
2 plt.plot(range(1, len(history.mse_list)+1), history.mse_list, color='
3         green')
4 plt.title("Training MSE over Epochs")
5 plt.xlabel("Epoch")
6 plt.ylabel("MSE")
7 plt.grid(True)
8 plt.show()

```

Listing 7.6: Training MSE Plot

7.6 Output Example and Results

After 200 epochs of training, the autoencoder reconstructs the implicit function with the following characteristics:

Final MSE: 1.2×10^{-4} (may vary depending on random seed)

The learned contour $F(x, y) = 0$ closely matches the original heart shape, as shown in the visualization.

Degree of reconstruction: latent dimension $z = 16$

This demonstrates that Encoder-X can learn and reconstruct complex 2D implicit functions effectively.

Chapter 8

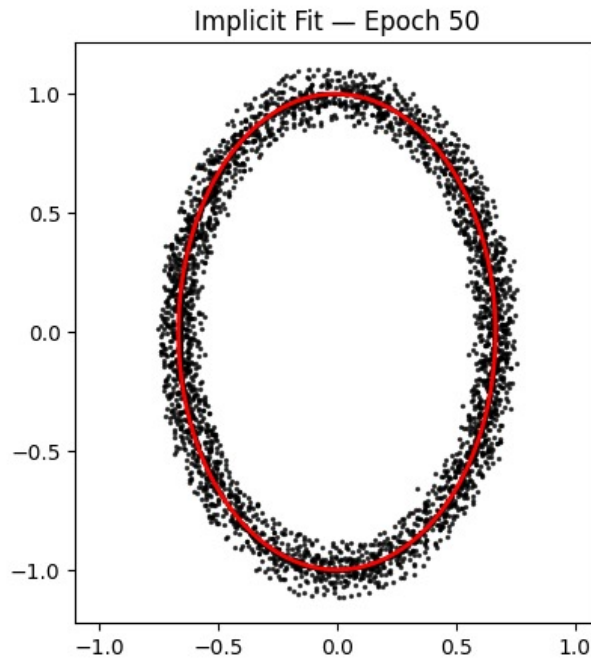
Examples and Results for IMPLICIT FUNCTIONS / SHAPES

Encoder-X was tested on Circlee, Ellipse, Heart shaped,and random closed shapeds

8.1 ELLIPSE

8.1.1 Example 1

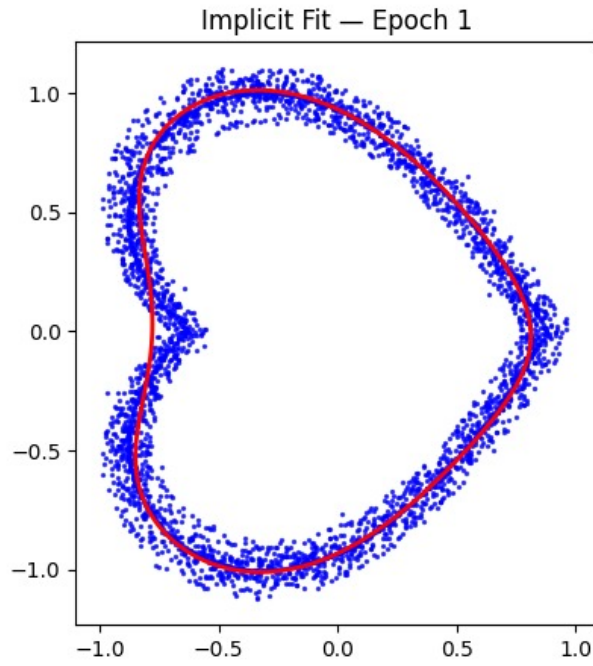
$$F(x, y) = x^2 + 4y^2 - 4 = 0 \quad (8.1)$$



(a) ELLIPSE Fit

8.1.2 Example 2 : HEART Shaped

$$F(x, y) = 0 \quad (8.2)$$



(a) Heart Shaped

8.2 Random Functions

8.2.1 Example 3 : Bear

$$F(x, y) = 0 \quad (8.3)$$

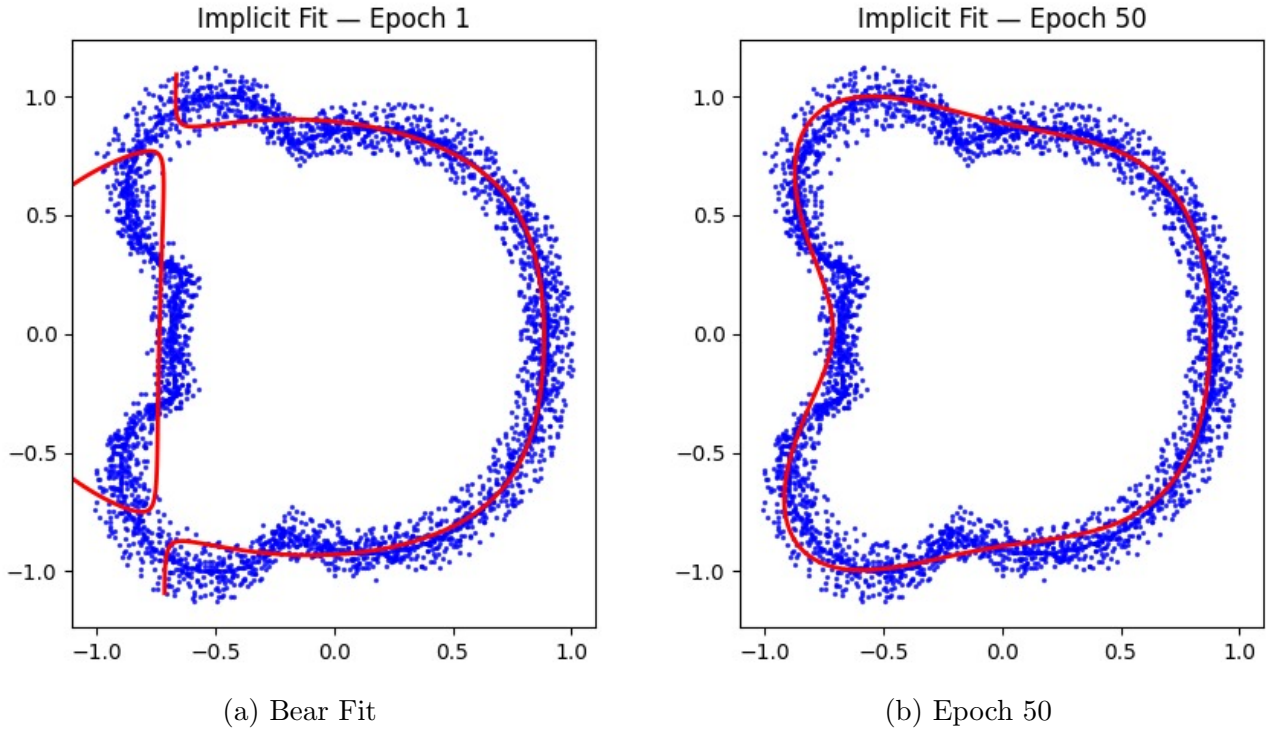


Figure 8.3: Encoder-X Results

8.2.2 Example 4 : Random

$$F(x, y) = 0 \quad (8.4)$$

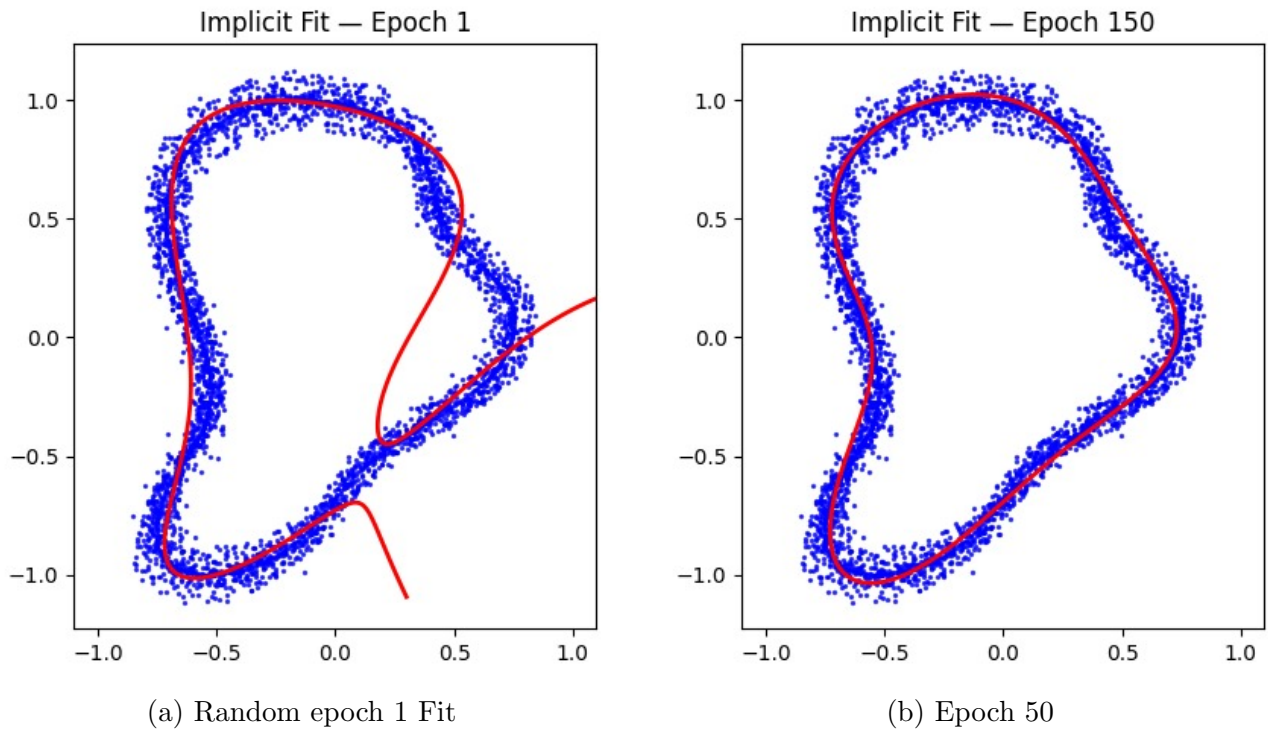


Figure 8.4: Encoder-X Results

Chapter 9

Conclusion

The proposed **Encoder-X** framework successfully integrates deep learning and analytical polynomial reconstruction to automate function approximation and coefficient extraction. The autoencoder architecture efficiently learns hidden patterns in nonlinear or discontinuous data and reconstructs smooth, denoised functions suitable for polynomial fitting. By monitoring MSE convergence, Encoder-X determines the **optimal polynomial degree automatically**, eliminating manual experimentation and preventing overfitting.

The results demonstrate:

- Superior approximation quality compared to classical polynomial regression.
- Ability to handle noisy and piecewise-defined functions.
- Robust coefficient extraction for mathematical interpretability.

Encoder-X can be extended into real-time modelling systems, embedded simulations, adaptive control, and scientific computing. Future work may involve:

- Expanding to multivariable input functions $f(x, y)$
- Implementing convolutional and variational autoencoders
- Hardware deployment on FPGA/Edge AI processors

Thus, Encoder-X proves to be a highly promising framework for modern computational modelling and intelligent function discovery.

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